

Kseniia Rozhkovaia

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Education

September 2023 – July 2027 – Bachelor's Degree in Sociology, Higher School of Economics, Russia

Major: "Sociology"

Minor "Intellectual Data Analysis"

GPA 8.96 / 10

Relevant coursework:

- Probability Theory & Mathematical Statistics I–III
- Algebra and Analysis
- Applied Software
- Quantitative Methods of Sociological Research I–II
- Data Analysis in Sociology I–III
- Introduction to Programming
- Demography
- Foundations of Machine Learning
- Foundations of Deep Learning
- Social Structure and Social Stratification
- Applied Data Analysis Problems
- Introduction to Data Platforms
- Analysis and Visualization of Data in R

August – December 2025 – Academic exchange, Tecnológico de Monterrey, Mexico

Relevant coursework:

- Introduction to Econometrics
- Business Programming
- Human Factors Research Methodologies
- Ethics and Artificial Intelligence

Academic Experience

Since April 2026 – Research Assistant, Centre for Comparative Research on Social Well-Being, HSE

- Conduct statistical analysis of large-scale nationally representative and cross-national survey datasets using Python and R for research on social values, attitudes, well-being, and cultural differences
- Apply multivariate statistical methods to identify patterns and associations between variables, construct respondent profiles, and analyse the structure of value orientations across social groups and countries
- Ongoing research: examining the relationship between attitudes toward success and respondents' basic human values measured with Schwartz's PVQ-21, including the analysis of success-related dimensions and value profiles using LCA and factor analysis

January – December 2026 – Teaching Assistant, Basics of Machine Learning & Deep Learning, Faculty of Computer Sciences, HSE

- Tutored 2 academic groups of 20 students each, answered questions on course material and assignments, provided additional learning resources, and graded homework
- Supported instructors with course delivery, student communication, and preparation of supplementary teaching materials

January – June 2025 – Teaching Assistant, Probability Theory and Mathematical Statistics, Faculty of Social Sciences, HSE

- Designed and graded examination papers
- Conducted regular consultation seminars for first-year students explaining statistical theory and problem-solving approaches

February – May 2024 – Intern, HSE Global Cities Innovation Index (GCII)-2024

- Acknowledged contributor, HSE Global Cities Innovation Index: 2024
DOI: 10.17323/978-5-7598-3023-8
- Researched and proposed potential indicators for the index, reviewing international conference databases, patent statistics, urban infrastructure data, technology company sources, and other publicly available information sources on urban innovation & technology
- Collected, cleaned and analysed data from heterogeneous sources and contributed to the refinement of the index's indicators framework
- Participated in the preparation of data visualisations and analytical materials for the final report

Work Experience

Since June 2026 – Data Analyst, Raiffeisen Bank Russia, CIB Trade Finance

- Develop process analytics for CIB Trade Finance, working with business-process and event-log data to measure the duration of key operational workflows, including credit-limit approval, guarantees, agreements, and letters of credit
- Designed and implemented an Approve-to-Sign process metric based on Camunda workflow logs, enabling measurement of the time between credit-limit approval and document signing across Trade Finance products
- Built a process analytics dashboard for key Trade Finance products, visualising end-to-end processing times based on the developed workflow-duration metric
- Develop data marts specifically for an LLM-based analytical agent, structuring business and process data to enable business teams to access process metrics and structured data insights through a user-friendly interface without recurring ad hoc requests to analysts

Since August 2023 – Teacher, Non-profit organisation “Association of Olympiad Winners”

- Teach online and in-person classes on Social Sciences & Economics for high school students
- Prepare talented students for participating in national academic competitions

Academic projects

Since September 2026, in progress – Bachelor’s thesis

“Large Language Models and Machine Learning for Modelling Human Values: From Measurement to Synthetic Population Generation”

- Design an ML-based algorithm to generate synthetic samples for human values surveys
- Examine whether persona-conditioned LLM responses can reproduce value profiles of different social and cultural groups and how cross-cultural differences are reflected in the models’ semantic representations
- Combine survey data analysis, statistical modelling and LLM-based experiments to evaluate the potential and limitations of language models as tools for sociological measurement

September 2025 – June 2026 – Term Paper

“The Role of Economic News on the Ruble Exchange Rate in Shaping Russians’ Perceptions of the Country’s Economic Situation”

- Developed data parsers for social networks and online news media using Python
- Analysed a corpus of approximately 220,000 economic news articles using BERTopic for topic modelling and RuBERT-based sentiment analysis to trace changes in media narratives and sentiment over time
- Integrated NLP-derived features with time-series data on the ruble exchange rate and a Consumer Sentiment Index constructed from nationally representative survey data, examining the relationship between media coverage, economic conditions and public perceptions

Languages

French – B2, DELF B2 (71.5/100), TCF TP (487/699)

English – C1, IELTS 8.0

Russian – Native

Technical Skills

Data Visualisation & Analytical Tools

Python (NumPy, SciPy, Pandas, Matplotlib, Seaborn), R, SPSS, Dashboard Development, Statistical Visualisation

Machine Learning & AI

Supervised and unsupervised learning, deep learning, NLP, transformer-based models, Scikit-learn, CatBoost, PyTorch, Hugging Face

Data Engineering & Databases

SQL; PostgreSQL, MySQL, Greenplum, ClickHouse; Apache Spark, PySpark, Hadoop, Hive, Trino, Apache Zeppelin; data marts, analytical data modelling, ETL/data preparation, ER and UML diagrams

Infrastructure & Development Tools

Git, GitHub, GitLab, Kubernetes, KubeFlow, Unix/Linux command line

Research & Quantitative Methods

Survey data analysis, cross-national and panel data analysis, multivariate statistical analysis, time-series analysis, statistical modelling, NLP-based text analysis, qualitative coding, research design, academic writing